

Web Development Notes

NAVTTTC Full Stack Development Classes - Arfa Karim Technology Incubator, Peshawar

1) Frontend (HTML, CSS, JS, React)

Ye wo cheez hai jo user dekhta hai. Jese kisi shop ki window display - colors, layout, buttons sab front-end hai. HTML structure deta hai (skeleton), CSS design/styling karta hai (colors, fonts), JS interactivity deta hai (button click pe kuch ho), React isi ko fast aur organized tareeqe se banane ka library hai.

2) Backend (Node, Express)

Ye wo hissa hai jo screen ke peeche kaam karta hai - jese shop ka stock room, jahan order process hota hai, payment check hoti hai. User ko dikhta nahi, lekin sab logic wahin chalta hai. Node.js JavaScript ko server pe chalata hai, Express us pe ek framework hai jo routes/APIs banana easy karta hai.

3) Database - NoSQL (MongoDB)

Data store karne ki jagah. Jese aap apni saari customer info ek register mein likho. SQL databases tables/rows mein data rakhte hain (jese Excel sheet), MongoDB documents (JSON jese) mein rakhta hai - flexible, structured tables ki zaroorat nahi.

4) Website

Web pages ka collection jo aapas mein links se connected hote hain. Jese ek school ki building mein bohat saare rooms hote hain - har room ek page, poori building website.

5) Browser -> Search Engine -> Internet

Browser = Chrome, Firefox (gari/vehicle jis se aap internet pe ghoomte ho). Search Engine = Google (rasta dhoondne wala guide). Internet = network of networks - duniya ke saare computers ek dosre se connected (jese global postal system jo har ghar tak letter pohanchata hai).

6-9) Network Types - LAN, MAN, WAN

LAN (Local Area Network) = ek ghar ya office ke andar computers connected (jese ghar ka WiFi). MAN (Metropolitan Area Network) = ek city ke andar network (jese Peshawar mein cable TV network jo poore city ko cover kare). WAN (Wide Area Network) = poori duniya ka network (jese Internet khud - Pakistan se America tak connection).

10) Domain

Website ka naam, jese google.com. Real life mein ye aapke ghar ka address hai - har koi us address pe aap tak pohanch sakta hai.

11) Hosting

Server storage jahan website ke files (HTML, CSS, images, database) rakhi jati hain. Jese aap ek dukaan kiraye pe lo taake apna saman rakh sako - hosting company aapko server space kiraye pe deti hai.

12) DNS (Domain Name System)

DNS domain name (jese thattheer.com) ko IP address (jese 192.16.8.x) mein convert karta hai, kyun ke computers naam nahi samajhte, numbers samajhte hain. Real life: jese aap kisi ka naam jaante ho lekin uska ghar dhoondne ke liye exact address (house number) chahiye - DNS yehi kaam karta hai, naam se address nikalta hai.

13) Client -> Request -> Server

Client = aap (browser se). Request = aap kuch maangte ho (jese mujhe homepage dikhao). Server = jo response deta hai. Real life: Restaurant mein aap (client) waiter ko order (request) dete ho, kitchen (server) khana bana ke deta hai.

14) SDLC (Software Development Life Cycle)

Software banane ka pura process/steps. Jese ghar banane ka bhi ek process hota hai - pehle plan, phir naksha, phir construction, phir check, phir maintenance.

15-17) SDLC ke Phases

i) Requirements - client se pucho usse kya chahiye (jese ghar banane se pehle pucho kitne rooms chahiye).
ii) Planning - kaise banayenge, kitna time/budget lagega. iii) Design vs Development vs Testing vs Deployment vs Maintenance: Design = naksha banana (UI/UX), Development = actual coding, Testing = check karna bugs hain ya nahi, Deployment = website ko live/launch karna (duniya ke liye available), Maintenance = baad mein updates/fixes karna (jese ghar ki regular repair).

18) Web Application - User Interact

Wo website jahan user sirf dekhta nahi, balke interact bhi karta hai - jese Facebook, Gmail. Aap data submit karte ho, login karte ho, comment karte ho.

19) Static vs Dynamic Website

Static = fixed content, sab users ko same dikhta hai (jese ek simple About Us page jo kabhi change nahi hota). Dynamic = content change hota hai based on user/data (jese Facebook feed, har user ka alag dikhta hai).

20) MERN vs MEAN

Dono full-stack combinations hain: MERN = MongoDB + Express + React + Node. MEAN = MongoDB + Express + Angular + Node. Farak sirf frontend framework ka hai - React vs Angular.

21) LMS (Learning Management System)

Online platform jahan courses/learning manage hota hai - jese Google Classroom ya Moodle, jahan teacher assignments dete hain aur students submit karte hain.

22) HTML Element Example

<p>hello Nabeel Documents Content</p> - Yahan <p> tag hai (paragraph), 'hello Nabeel Documents Content' uska content hai, aur poora <p>...</p> combination ko element kehte hain. Jese ek box (tag) ke andar cheez (content) rakhi ho - box + cheez = element.